Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer specializing in **AI agent systems and agentic AI evaluation** - built AgentProbe, an open-source multi-agent toolkit with 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix), async Python/FastAPI backend, LLM-powered eval engine, and automated prompt iteration. 8+ years production backend engineering at scale (1M+ records, 100K+ MAU, Python/TypeScript/AWS). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Multi-Agent Orchestration | LangChain / LangGraph | Async Python & TypeScript | Tool Calls & Agent Reasoning | LLM Evaluation & Judges | RAG Pipelines / ChromaDB | FastAPI / REST APIs | Enterprise AI Deployment | Production Debugging | AWS Distributed Pipelines | PostgreSQL / SQL Tuning | Docker / Microservices

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **20+ full-stack production modules** (Python, TypeScript, PostgreSQL) across **120+ Agile sprints** - reducing deployment times by **40%** with cross-functional teams for **100K+ monthly active users**.
- Cut microservices execution time by **50%** by leading Docker-based containerization for 30%+ of modules, delivering scalable async backend architecture supporting **1M+ property records**.
- Resolved **100+ high-priority bugs** and built 5+ debugging tools reducing resolution time by **40%** - maintaining **99.5% uptime** and sub-10 bug count for 6+ consecutive months via TDD/BDD gates.
- Improved production query performance by **30%** by architecting PostgreSQL schemas, data access layers, and indexing strategies across 20+ projects.
- Achieved **100% PHI compliance** migrating all client data to AWS S3 with automated pipelines, Libsodium encryption, and zero security incidents - eliminating 80% of manual workload.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates through sprint-style feedback cycles.

PROJECTS

AgentProbe: Multi-Agent AI Evaluation Toolkit OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure · TypeScript

Built a **production-grade multi-agent system** for evaluating and improving agentic AI pipelines. Architecture: 3 async LangGraph agents - **Orchestrator** (routes queries, manages tool calls), **RAG Tracer** (trace enrichment per agent run), **Auto Fix** (automated prompt iteration when eval scores degrade). Smart LLM Router for dynamic model routing, **15-endpoint FastAPI async backend**, React dashboard with per-query agent trace visualization. LLM-powered judges score 5 metrics: correctness, completeness, hallucination, retrieval quality, compliance. Supports Ollama, OpenAI, Anthropic, Groq - runs 100% locally at zero cost. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Built NLP pipeline mapping natural language queries (EN/ES) to FHIR APIs via LLMs, BERT, and embeddings. HIPAA-compliant RBAC, containerized. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: UK Road Accident Analysis

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Predictive modeling on 100K+ accident records. **85%+ accuracy**; interactive GitHub Pages dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS - async programming across all backend layers

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, Tool Calls, LLM Judges, ChromaDB, RAG, Embeddings, Ollama, OpenAI, Anthropic, Groq, FastAPI, Prompt Engineering

Databases: PostgreSQL, MySQL, Oracle, NoSQL, GraphQL - schema design, query tuning, indexing

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & NLP: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Feature Engineering, Predictive Modeling

Testing: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum